AMPHORA TECHNICAL ENVELOPE — PROJECT "SUBSTRATE"

Discovery and Iteration Envelope: Manufacturable Graphene-Alternative 2D Membrane/Additive Platform

Envelope class: Parent discovery envelope with three child application tracks **Posture:** Material-agnostic. Do not pre-commit to a candidate class. Converge via scoring function (§6). **Prime directive:** Identify, validate, and iterate toward a 2D or quasi-2D nanomaterial platform that delivers ≥70% of graphene's relevant functional performance per application track at ≤5% of graphene's at-scale production cost, using synthesis routes compatible with existing industrial infrastructure (solution processing, roll-to-roll, molten-salt electrochemistry, flash Joule heating, or CVD-free exfoliation).

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§1 — MISSION CONTEXT

Graphene's barrier is not performance — it is manufacturability: high-quality monolayer CVD graphene runs $60k–$200k/kg equivalent; even bulk graphene nanoplatelets with acceptable defect density exceed viable cost for commodity applications (concrete, alloys) and struggle with batch consistency for precision applications (ion-selective membranes). The envelope's objective is a substitute platform — one material class or a small engineered family — that is:

1. Synthesizable from abundant precursors (target feedstock cost <$2/kg)
2. **Feedstock and critical process inputs sourceable within North America (US/Canada/Mexico)** — no supply-chain dependency on precursors with single-region (esp. non-NA) production concentration
3. Scalable to tonne-per-year within 24 months of process lock
4. Functionalizable for all three application tracks from a common base process
5. Free-to-operate or licensable on acceptable terms (see §3.4)

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§2 — CANDIDATE SPACE (SEED, NOT CONSTRAINT)

The discovery layer must rank the full known 2D/nanomaterial space. Seed classes for initial index construction only — the system is expected to expand this list from literature ingestion:

- MXenes ($Ti_3C_2T_x$ and the broader $M_{n+1}X_nT_x$ family; molten-salt and acid-free synthesis routes)
- Transition metal dichalcogenides (MoS_2, WS_2, $MoSe_2$)
- Hexagonal boron nitride (hBN) and BN nanosheets
- Graphyne / graphdiyne
- Borophene and borophane
- Phosphorene / black phosphorus
- Graphitic carbon nitride (g-C_3N_4)
- Turbostratic "flash" graphene (flash Joule heating from waste carbon — treat as a graphene-cost-disruptor candidate, not excluded by the graphene name)
- Reduced graphene oxide and holey GO (low-cost, defect-tolerant applications)
- Layered double hydroxides (LDH)
- Silicene, germanene, stanene
- Clay-derived and geopolymer nanosheets (kaolinite, montmorillonite exfoliates)
- 2D zeolites and 2D MOFs/COFs (membrane track especially)
- Boron carbide and SiC nanosheets

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§3 — PHASE 0: DISCOVERY, INGESTION, PRIOR-ART MAPPING

Spawn sub-repo `substrate-discovery` before any application-track work. It owns four pipelines:

3.1 Academic literature ingestion

- Sources: arXiv (cond-mat.mtrl-sci), ChemRxiv, Nature/Science/ACS/RSC/Elsevier via licensed API access where held, Semantic Scholar API, OpenAlex, CORE.
- Dissertations: ProQuest Dissertations & Theses, EThOS (British Library), DART-Europe, HathiTrust, university institutional repositories (MIT DSpace, Stanford, Berkeley eScholarship, Tsinghua/KAIST/NUS open repositories).
- Extraction targets per document: material system, synthesis route, precursor cost class, reported properties (with units normalized to envelope schema), scale of demonstration (mg / g / kg), stability data, and named failure modes.
- Build a normalized properties knowledge graph: `material → synthesis_route → property_vector → demonstrated_scale → citation`.

3.2 Computational materials databases

- Materials Project, AFLOW, OQMD, NOMAD, 2DMatPedia, C2DB (Computational 2D Materials Database), JARVIS-DFT.

- Pull formation energies, exfoliation energies, band structures, elastic tensors, and ion-migration barriers where computed. Flag candidates with exfoliation energy <150 meV/atom as solution-processing-viable.

3.3 IP landscape (prior-art and freedom-to-operate mapping)

- Sources: USPTO PatFT/AppFT and PEDS, EPO Espacenet/OPS API, WIPO PATENTSCOPE, Google Patents Public Datasets (BigQuery), Lens.org, CNIPA for Chinese filings (dominant in MXene and GO space).
- **Use official APIs and bulk-data products only; respect rate limits and terms of service. This pipeline performs prior-art mining and FTO landscaping — it does not reproduce or ingest paywalled full-text without license.**
- Deliverables: (a) claim-density heatmap per candidate class per jurisdiction; (b) expiry timeline — flag foundational patents expiring within 36 months; (c) white-space report — property/application combinations with thin claim coverage; (d) assignee concentration index — candidates dominated by a single hostile assignee score down in §6.
- Route the FTO output to human counsel before any process lock. The system generates the landscape; it does not render legal opinions.

3.4 Cost-of-manufacture modeling

- For every candidate surviving initial screen, construct a bottom-up cost model: precursor $/kg, energy intensity (kWh/kg), yield, capex class of required equipment, batch vs continuous.
- Benchmark column: same model run for CVD graphene, GNP, and GO as reference anchors.

3.5 Feedstock supply-chain screen (North America)

- For every precursor and critical process input, map: producing mines/plants in US/Canada/Mexico, current NA annual capacity vs projected demand at tonne-scale, number of independent NA suppliers, and USGS critical-minerals status.
- Data sources: USGS Mineral Commodity Summaries, NRCan mineral statistics, EIA, S&P Capital IQ commodity data where licensed.
- Classification per candidate: **NA-secure** (≥2 independent NA suppliers at required volume) / **NA-possible** (NA reserves exist but capacity buildout required — flag lead time) / **NA-blocked** (structurally import-dependent, e.g., precursors with >80% single-country foreign concentration).
- NA-blocked is a hard gate (§6). NA-possible candidates carry a supply-risk discount and must include a capacity-buildout note in the cost model.
- Example implications for the seed space: Ti/Al/C for MXenes, borates (major CA/US production) for BN and borophene classes, and waste-carbon feedstock for flash graphene are NA-favorable; black phosphorus and several TMD precursors (Mo is

NA-viable; some Se/W supply chains are not) require explicit mapping — verify, don't assume.

Phase 0 exit criterion: ranked shortlist of ≤8 candidate systems with property vectors, cost models, and IP posture. Shortlist triggers child-track sub-repo creation (§5).

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§4 — CHILD APPLICATION TRACKS

Each track inherits the Phase 0 knowledge graph and shortlist. Tracks run in parallel; a candidate may live in multiple tracks. Track-specific KPIs below are pass/fail gates for iteration promotion.

Track A — `substrate-dle`: Direct Lithium Extraction Membrane

- **Function:** Li^+-selective transport membrane for brine and geothermal sources.
- **KPIs:**
  - Li^+/Mg^{2+} selectivity ≥ 50 (stretch: 200) — the governing figure of merit for Salar-type brines
  - Li^+ permeance ≥ 0.1 mol m^{-2} h^{-1} bar^{-1} at operating conditions
  - Stability: <10% flux decline over 1,000 h in synthetic brine (pH 4–10, up to 300 g/L TDS)
  - Fouling resistance and cleanability (CIP-compatible chemistry)
  - Membrane cost ≤ $50/$m^2$ at scale
- **Iteration levers:** interlayer spacing control, surface charge/termination chemistry, crown-ether or sulfonate functionalization, laminate vs mixed-matrix architectures, support-layer selection.
- **Simulation stack:** MD ion transport (LAMMPS/GROMACS), DFT for binding energies at pore terminations, continuum transport models for module-scale projection.

Track B — `substrate-concrete`: Cementitious Strength and Longevity Additive

- **Function:** dispersible nanosheet admixture for compressive/flexural gain and durability.
- **KPIs (vs control mix, standard ASTM/EN test matrix):**
  - Compressive strength: +15% at 28 days (ASTM C39) at dosage ≤0.1 wt% of cement
  - Flexural strength: +20% (ASTM C78)
  - Chloride ingress reduction ≥40% (ASTM C1202 / NT Build 492)
  - Freeze-thaw durability factor ≥90 after 300 cycles (ASTM C666)
  - No set-time or workability violation (ASTM C143 slump within spec, C191 set time)

    - Additive cost contribution ≤ $8/m³ of concrete
- **Iteration levers:** dispersion chemistry (surfactant/superplasticizer compatibility — polycarboxylate ether interaction is a known failure mode), sheet lateral size vs C-S-H nucleation, surface oxygen content, dry-dispersible vs slurry delivery format.
- **Note:** dispersion at scale, not intrinsic strength, is the historical failure mode of graphene-in-concrete. Weight the dispersion sub-problem accordingly.

Track C — `substrate-alloy`: Aluminum Matrix Reinforcement

- **Function:** nanosheet reinforcement for Al and Al-alloy MMCs (target series: 5xxx, 6xxx, 7xxx).
- **KPIs (vs unreinforced baseline, same temper):**
    - Yield strength +25%, UTS +15% at ≤2 vol% loading
    - Elongation retention ≥80% of baseline (no catastrophic ductility loss)
    - Fatigue life ≥1.5× at 10^7 cycles
    - Corrosion: no degradation vs baseline in ASTM B117 salt spray (galvanic coupling with carbon fillers is a known killer — score chemically inert candidates, e.g., BN-class, up)
    - Process compatibility: stir casting or powder metallurgy route; no requirement for SPS-only processing
- **Iteration levers:** wettability engineering (coatings, in-situ reaction layers), dispersion in melt vs solid-state routes, interface reaction products (Al_4C_3 formation is a failure mode for carbon-based fillers — model interfacial thermodynamics explicitly).

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§5 — SUB-REPO SPAWNING AND ITERATION PROTOCOL

- **Naming:** `substrate-{track}-{candidate}` (e.g., `substrate-dle-mxene-ti3c2`).
- **Spawn criterion:** candidate passes Phase 0 screen for that track's property prerequisites AND cost model projects ≤5% of graphene reference AND no red-flag FTO block.
- **Kill criterion:** two consecutive iteration cycles with no KPI improvement, or cost model revision pushing above 10% of graphene reference, or FTO downgrade to blocked. Killed repos archive their knowledge-graph deltas back to `substrate-discovery`.
- **Iteration loop per sub-repo:**
    1. Hypothesis generation from knowledge graph (composition, termination, architecture)
    2. Computational screen (DFT → MD → mesoscale as appropriate to track)
    3. Synthesis-route feasibility check against cost model
    4. Rank against §6 scoring function; log all results including failures to the shared graph

5. Promote top decile to physical-validation queue (human lab handoff spec: full synthesis protocol, characterization checklist, expected property window with uncertainty bounds)

- **Cross-track arbitration:** weekly merge cycle where `substrate-discovery` re-ranks the global candidate list using new evidence from all tracks. A candidate winning in two or more tracks receives a platform bonus (§6).

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§6 — SCORING FUNCTION

Composite score per candidate per track, normalized 0–100:

Component	**Weight**
Performance vs track KPIs (graphene = reference 100)	0.30
Projected at-scale cost ($/kg, log-scaled vs graphene anchor)	0.30
Manufacturability (TRL of synthesis route, equipment capex class, batch consistency evidence)	0.15
Feedstock security (NA-secure = full marks; NA-possible discounted by buildout lead time)	0.10
IP posture (white space > expiring > licensable > concentrated hostile)	0.08
Platform breadth (bonus for multi-track viability from common base process)	0.07

Hard gates (score irrelevant if failed): **NA-blocked feedstock classification (§3.5)**; environmental/regulatory viability of synthesis chemistry (e.g., HF-based MXene etching scores as a route-risk unless an acid-free alternative is in the same repo); and demonstrated stability under track operating conditions.

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§7 — OUTPUT ARTIFACTS

Per iteration cycle, each active sub-repo emits:

1. Ranked candidate ledger with full property vectors and provenance citations
2. Cost model delta report
3. FTO status delta
4. Physical-validation handoff specs for promoted candidates
5. Failure log (mandatory — negative results feed the shared graph)

Parent envelope emits monthly: global convergence report, kill/spawn log, and a single-page executive summary (performance-vs-cost frontier plot, all candidates, all tracks).

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§8 — GUARDRAILS

- All external data acquisition through official APIs, licensed access, or open repositories. No circumvention of paywalls or robots directives.
- IP outputs are landscaping inputs for human counsel, not legal conclusions.
- All application tracks in this envelope are civilian-industrial (mining, construction, metallurgy). Any proposed extension outside these tracks requires a new envelope with human sign-off.
- Physical synthesis and testing remain human-executed; the system's authority ends at the validation handoff spec.